Problem Set 5

Exercise 1

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Setup

- First, we estimate the model:

$$y = \alpha_0 + \alpha_1 sex + \alpha_2 dex + \alpha_3 lex + \alpha_4 lex^2 + u$$

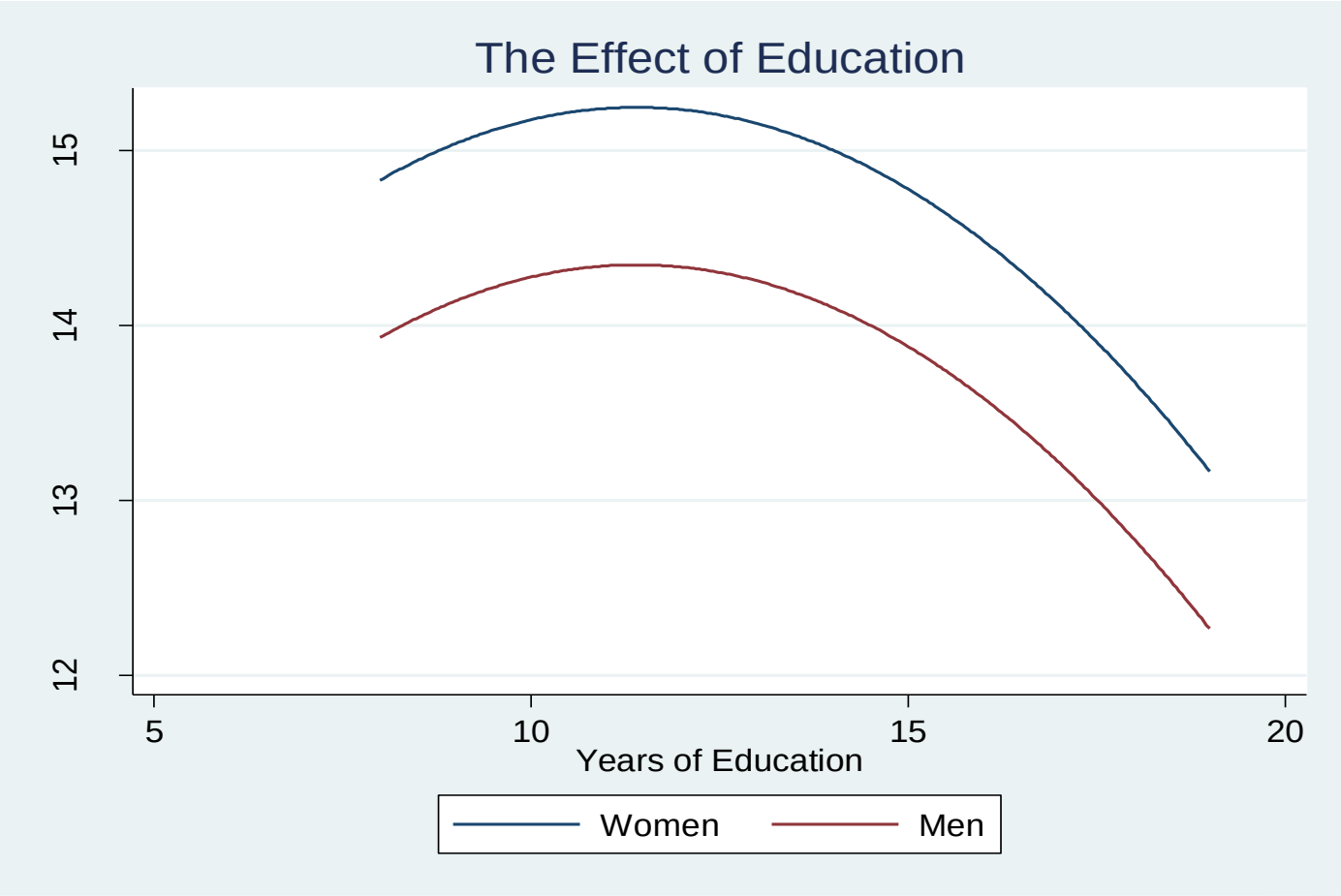
```
Residuals:
    Min      1Q  Median      3Q     Max 
-3.9094 -0.7227  0.0357  0.7647  3.1867 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  5.527936   2.031914   2.721  0.00668 ** 
sex         -0.900641   0.087489 -10.294  < 2e-16 ***
dex          0.112088   0.006003  18.671  < 2e-16 ***
lex          0.821391   0.321307   2.556  0.01079 *  
lex2        -0.036032   0.012808  -2.813  0.00505 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.124 on 678 degrees of freedom
Multiple R-squared: 0.3883,	Adjusted R-squared: 0.3847 
F-statistic: 107.6 on 4 and 678 DF,  p-value: < 2.2e-16
```

Question (a)

- To test: H_0: $\alpha_3 = \alpha_4 = 0$
  - F-Test:
    - R: $y = \alpha_0 + \alpha_1 sex + \alpha_2 dex + u$
    - U: $y = \alpha_0 + \alpha_1 sex + \alpha_2 dex + \alpha_3 lex + \alpha_4 lex^2 + u$
    - $\frac{(RSS_R - RSS_U)/q}{RSS_U/(n-k)} \sim F_{q,n-k}$
    - F-Statistic: 6.0613 and p-value: 0.00246 → Reject H_0
- To test: H_0: $\alpha_4 = 0$
  - Reject H_0

Question (b)

- Dependence of mean productivity on education, by gender



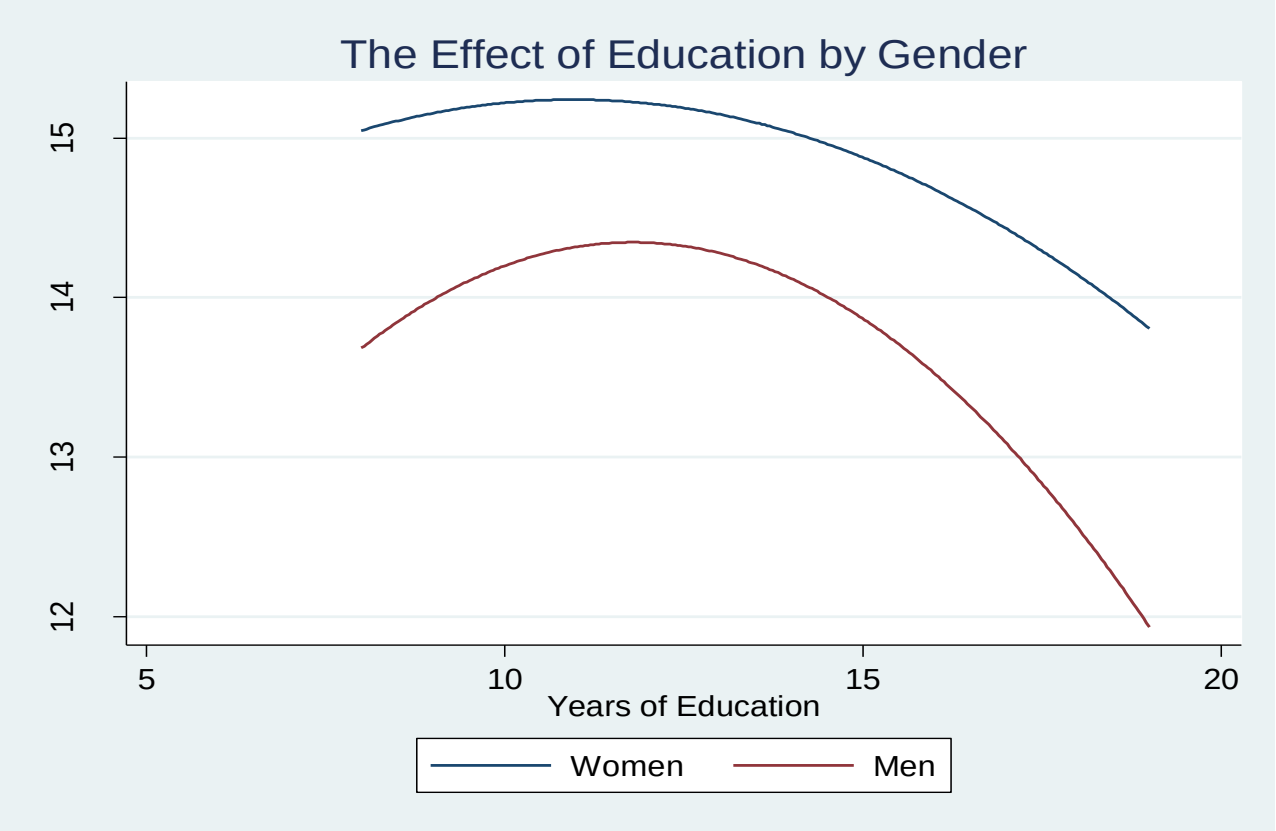
Question (b)

- Can the whole shape of the education effect be different?
- Now, we add interaction terms and estimate the model:

$$y = \alpha_0 + \alpha_1 sex + \alpha_2 dex + \alpha_3 lex + \alpha_4 lex^2 + \alpha_5 lex * sex + \alpha_6 lex^2 * sex + u$$

```
Residuals:
    Min      1Q  Median      3Q     Max 
-3.8937 -0.7313  0.0499  0.7602  3.1874 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  7.532590   3.128583   2.408   0.0163 *  
sex         -4.668065   4.305661  -1.084   0.2787    
dex          0.112284   0.006021  18.647   <2e-16 ***
lex          0.485620   0.522063   0.930   0.3526    
lex2        -0.022157   0.021878  -1.013   0.3115    
sex:lex      0.606303   0.693945   0.874   0.3826    
sex:lex2    -0.024168   0.028086  -0.861   0.3898    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.125 on 676 degrees of freedom
Multiple R-squared: 0.389,	Adjusted R-squared: 0.3835 
F-statistic: 71.72 on 6 and 676 DF,  p-value: < 2.2e-16
```

Question (b)



To test if shapes of the education effect are different for men and women:

H_0: $\alpha_5 = \alpha_6 = 0$ F-Test: 0.3836 and p-value: 0.6815 → Cannot reject H_0

Question (c)

- We maximize productivity at:

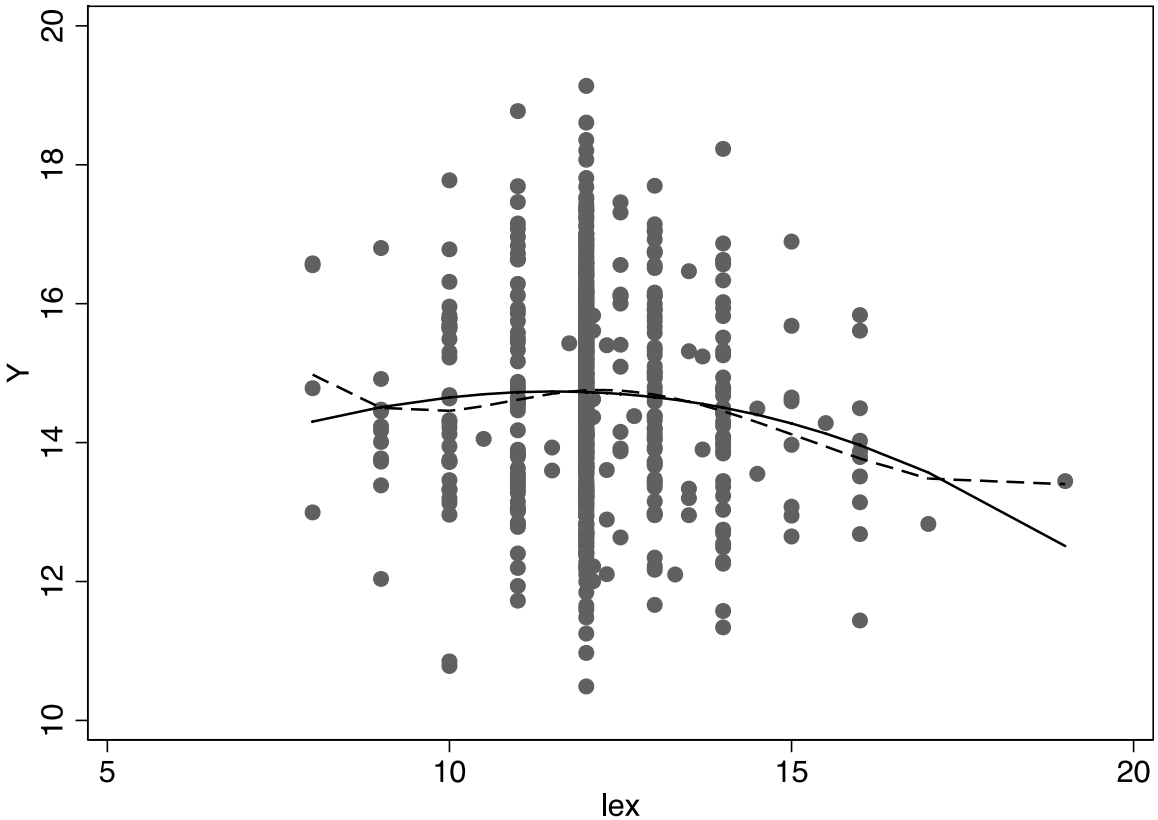
$$lex^* = -\frac{\alpha_1}{2\alpha_2}$$

- Bootstrap:
  - 1,000 repetitions
  - lex* = 11.4 $CI_{95\%}$=[5.70, 17.09]

- Delta-Method:
  - lex* = 11.4 $CI_{95\%}$=[10.16, 12.64]

Question (d)

- Cubic B-spline for education:



Note: solid line – quadratic fit; dashed line – cubic b-spline fit with knots(8 12 19)